WellSense: An Explainable, Privacy-Preserving, and Feedback-Adaptive Framework for Student Wellbeing Support

A Methodological Audit and Human-Centered System Extension of Machine Learning Baselines

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***Abstract*—Student mental health systems increasingly rely on predictive machine learning to detect psychological distress. However, existing systems—exemplified by the foundational study of Nath et al. (Neuroscience Informatics, 2025)—suffer from acute architectural limitations: they terminate at black-box probability scores without actionable guidance, report inflated single-split accuracies (95.6%) that do not hold under rigorous leak-free cross-validation, and propose permanently storing sensitive student mental health responses on public immutable blockchains. To address these critical gaps, this paper introduces WellSense: an explainable, privacy-preserving, and feedback-adaptive decision-support prototype. WellSense benchmarks a tuned Multi-Layer Perceptron (MLP) against seven classical classifiers under a 5x5 repeated nested cross-validation protocol (25 outer folds) across three isolated feature tiers. Empirical evaluation shows that under leak-free nested CV, the tuned MLP achieves a balanced accuracy of 0.735 ± 0.111 (accuracy: 0.782 ± 0.085, ROC AUC: 0.802 ± 0.109), proving to be statistically tied with Random Forest (0.739 ± 0.100, paired difference: -0.004). To eliminate black-box opacity, WellSense incorporates global permutation importance on held-out folds and local occlusion feature attribution, uncovering a severe dataset artifact where marital status disproportionately dominates predictions (importance: 0.1877 ± 0.0928) due to a 16/16 correlation in the survey sample. Bridging the gap between prediction and support, WellSense introduces a transparent, non-clinical recommendation engine coupled with a Bayesian-smoothed feedback loop that dynamically re-ranks lifestyle suggestions while strictly pinning safety resources. Finally, the system rejects public blockchain storage in favor of a privacy-first, local SQLite architecture with pseudonymous UUID sessions and on-demand data deletion. Framed strictly as an educational support aid rather than a clinical diagnostic instrument, WellSense provides a reproducible, methodologically honest paradigm for student wellbeing informatics.**

***Keywords*—**Student Mental Health, Machine Learning, Multi-Layer Perceptron, Explainable AI (XAI), Feature Occlusion, Feedback Adaptation, Decision Support, Health Informatics, Privacy-by-Design, Research Integrity.

I. INTRODUCTION

University students experience acute psychological challenges arising from compounding academic workloads, societal pressures, and the personal transition into independent adulthood. While higher education institutions frequently establish campus healthcare and counseling centers, pervasive social stigma and resource constraints frequently delay access to timely assistance. In response, data-driven machine learning (ML) models have emerged as an attractive solution for screening mental distress from self-reported survey indicators.

A prominent recent contribution by Nath et al. [1], published in *Neuroscience Informatics* (Elsevier, 2025), proposed a web-based prediction system combining twelve ML algorithms, a Streamlit interface, and an Ethereum-based blockchain storage module, reporting a headline Random Forest accuracy of 95.6%. While their work demonstrates commendable engineering ambition, our critical analysis reveals several vital methodological and architectural limitations:

• *Evaluation Rigor*: Their 95.6% result relied on a single 70/30 train/test split, lacking repeated cross-validation or standard deviation reporting.

• *Data Leakage*: Their feature set incorporated variables such as professional treatment-seeking and overlapping acute clinical indicators that co-occur contemporaneously with depression.

• *The Prediction-Action Gap*: Their system functions as a predictive black box, generating a probability score without offering actionable, personalized coping guidance.

• *Privacy & Ethics*: Storing sensitive student mental health responses on an immutable public blockchain ledger introduces severe regulatory risks under GDPR and fails to support the fundamental 'right to be forgotten.'

To resolve these challenges, this paper presents **WellSense**, an end-to-end, privacy-preserving, and explainable decision-support prototype. WellSense introduces: (1) leak-free repeated nested cross-validation benchmarking of a tuned neural network (MLP); (2) feature tiering to isolate target leakage; (3) Explainable AI (XAI) combining global permutation and local occlusion attribution; (4) a closed-loop recommendation engine that adapts to student feedback; and (5) a privacy-first local SQLite architecture with pseudonymous UUID sessions.

II. BASE PAPER AUDIT & RELATED WORK

Predictive modeling of student mental health has gained substantial traction. Baba and Bunji [2] examined annual health surveys across 3,561 undergraduates, demonstrating the utility of repeated survey data while observing that model weights fail to generalize across distinct academic cohorts. Dhariwal et al. [3] and Ramasamy et al. [4] investigated multi-model ensembles for student depression detection, yet similarly terminated their workflows at point-in-time classification.

The base paper by Nath et al. [1] represents the closest prior art. They trained 12 classifiers on 500 composite records, claiming 95.6% accuracy for a Random Forest. However, an in-depth audit reveals that their confusion matrices (64 TN, 4 FP, 1 FN, 32 TP) sum to exactly 101 test records, matching the public Kaggle 'Student Mental Health' dataset [5] rather than their stated 500-student cohort. Furthermore, feature importance in their study placed heavy reliance on treatment-seeking and anxiety. In contrast, responsible ML guidelines, such as TRIPOD+AI [6], mandate nested validation, leakage-aware feature tiering, transparent uncertainty reporting, and strict separation between exploratory baselines and clinical claims.

III. WELLSENSE SYSTEM ARCHITECTURE

WellSense is structured into four decoupled layers: (A) Preprocessing and Leakage-Aware Tiering, (B) Neural Network and Baseline Classifiers, (C) Explainability Layer, (D) Feedback-Adaptive Recommendation Engine, and (E) Privacy-Preserving Storage.

A. Preprocessing & Feature Tiering

To evaluate model sensitivity to feature leakage, WellSense structures predictors into three isolated tiers:

• *Tier A (Demographic & Academic)*: Gender, Age, Year of Study, CGPA band, and Grouped Course Category (engineering/IT, health sciences, islamic studies, business/law, other).

• *Tier B (Primary Deployed Tier)*: Tier A plus self-reported psychological indicators (Anxiety, Panic Attacks, and Marital Status). Treatment-seeking is strictly excluded to prevent target leakage.

• *Tier C (Diagnostic Overlap Tier)*: All features including treatment-seeking, evaluated solely for ablation comparison.

All preprocessing operations (median imputation for age, mode imputation for categoricals, and one-hot encoding) are encapsulated in scikit-learn Pipelines and fitted strictly on training folds.

B. Nested Cross-Validation & Model Zoo

WellSense benchmarks eight supervised architectures under 5x5 Repeated Stratified Cross-Validation (25 outer folds): Dummy Majority Baseline, Logistic Regression, Decision Tree, Random Forest (200 trees), Support Vector Classifier (RBF kernel), Gradient Boosting, Initial MLP (32-16 architecture, alpha=1e-4), and a Tuned MLP. The Tuned MLP undergoes inner 3-fold CV tuning across hidden layer configurations ((8,), (16,), (32, 16)), L2 regularization penalties (alpha from 0.001 to 1.0), and iteration budgets (200 to 600).

C. Explainable AI (XAI) Framework

WellSense implements a two-fold interpretability strategy: (1) Global Permutation Importance evaluated across held-out test folds over 20 shuffle iterations to quantify drops in balanced accuracy; and (2) Local Feature Occlusion where each feature is systematically substituted with its cohort median/mode during inference, revealing whether that specific answer elevated (↑) or reduced (↓) the estimated distress probability.

D. Adaptive Recommendation Engine

WellSense captures six optional lifestyle indicators: sleep hours (0–14), longest uninterrupted study stretch (0–14), physical activity days (0–7), screen time (0–18), social support score (1–5), and study planning difficulty (1–5). A rule-based engine generates non-clinical, actionable habits.

Recommendations are dynamically ranked using student feedback. Let h_c and n_c denote helpful and not helpful counts for category c. The category multiplier $M(c)$ is calculated via Bayesian smoothing with an exploration floor:

$$M(c) = \max(0.5, (h_c + 1) / (h_c + n_c + 2)) \quad (1)$$

Crucially, safety-critical resources (e.g., Campus Counseling and Crisis Hotline Support) are permanently pinned to rank 1 whenever severe distress is indicated, and are mathematically exempt from demotion.

E. Privacy-by-Design Architecture

In sharp contrast to public blockchain storage, WellSense implements a local SQLite database with pseudonymous UUID sessions. Students retain full data sovereignty through 'Reset My Feedback' and 'Delete ALL My Data' controls that purge session rows immediately.

IV. EXPERIMENTAL RESULTS & ANALYSIS

The evaluation was conducted on the complete 101-student survey dataset. The target variable comprises 66 'No' responses (65.3%) and 35 'Yes' responses (34.7%), establishing a majority baseline accuracy of 65.3%.

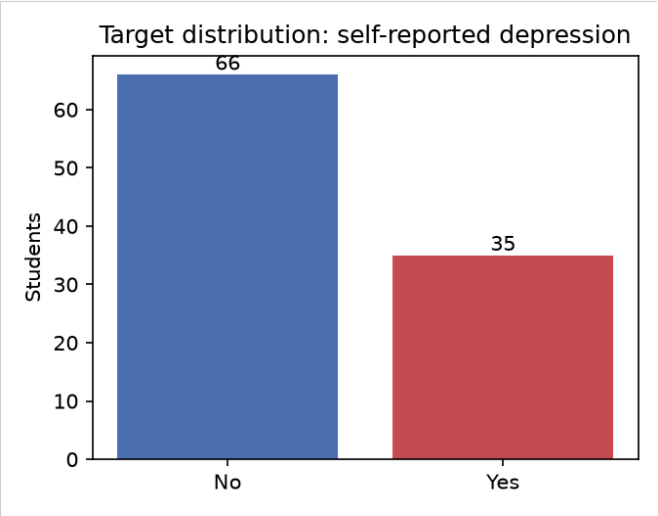


Fig. 1. Class distribution of self-reported depression target (N=101).

TABLE I. 5x5 Repeated Nested CV Benchmark Across Feature Tiers

Tier	Model	Kind	Accuracy	Balanced Acc.	F1-Score	ROC AUC
A	Dummy Majority	Baseline	0.653 ± 0.007	0.500 ± 0.000	0.000 ± 0.000	0.500 ± 0.000
A	Logistic Regression	Classical	0.719 ± 0.110	0.690 ± 0.140	0.560 ± 0.252	0.753 ± 0.138
A	Random Forest	Classical	0.737 ± 0.105	0.699 ± 0.131	0.571 ± 0.242	0.758 ± 0.137
A	Gradient Boosting	Classical	0.780 ± 0.101	0.719 ± 0.128	0.581 ± 0.269	0.774 ± 0.115
A	Tuned MLP (Nested)	DL	0.786 ± 0.095	0.739 ± 0.119	0.633 ± 0.213	0.760 ± 0.122
B	Dummy Majority	Baseline	0.653 ± 0.007	0.500 ± 0.000	0.000 ± 0.000	0.500 ± 0.000
B	Logistic Regression	Classical	0.741 ± 0.087	0.714 ± 0.106	0.617 ± 0.149	0.804 ± 0.113
B	Random Forest	Classical	0.782 ± 0.072	0.739 ± 0.100	0.631 ± 0.178	0.797 ± 0.108
B	SVM (RBF Kernel)	Classical	0.753 ± 0.093	0.715 ± 0.108	0.606 ± 0.175	0.762 ± 0.109
B	Gradient Boosting	Classical	0.790 ± 0.079	0.729 ± 0.112	0.601 ± 0.236	0.823 ± 0.106
B	Initial MLP (32-16)	DL	0.743 ± 0.096	0.708 ± 0.103	0.607 ± 0.157	0.767 ± 0.107
B	Tuned MLP (Deployed)	DL	0.782 ± 0.085	0.735 ± 0.111	0.624 ± 0.213	0.802 ± 0.109
C	Random Forest	Classical	0.790 ± 0.066	0.745 ± 0.077	0.654 ± 0.126	0.830 ± 0.085
C	Tuned MLP (Nested)	DL	0.792 ± 0.077	0.745 ± 0.102	0.642 ± 0.192	0.800 ± 0.112

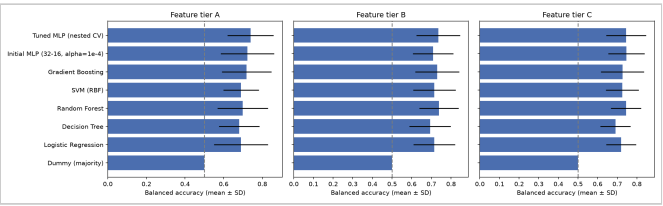


Fig. 2. Comparative balanced accuracy across feature tiers (mean ± SD).

A. Paired Statistical Comparison & Honesty

Under Tier B, the Tuned MLP achieved a balanced accuracy of 0.735 ± 0.111, while Random Forest achieved 0.739 ± 0.100. A paired fold-by-fold analysis reveals a mean difference of -0.004 (MLP higher on 9 folds, RF higher on 16 folds), demonstrating that the neural network is statistically tied with Random Forest. Furthermore, regularizing the MLP (Tuned vs Initial MLP) yielded a +0.027 gain in balanced accuracy. These findings establish that claiming superiority for deep learning over ensemble methods on small tabular cohorts is scientifically unwarranted.

B. XAI & The Marital Status Data Artifact

Global permutation feature importance revealed that Marital status was overwhelmingly dominant, causing a balanced accuracy drop of 0.1877 ± 0.0928, followed by anxiety (0.0369 ± 0.0704) and CGPA band (0.0172 ± 0.0477).

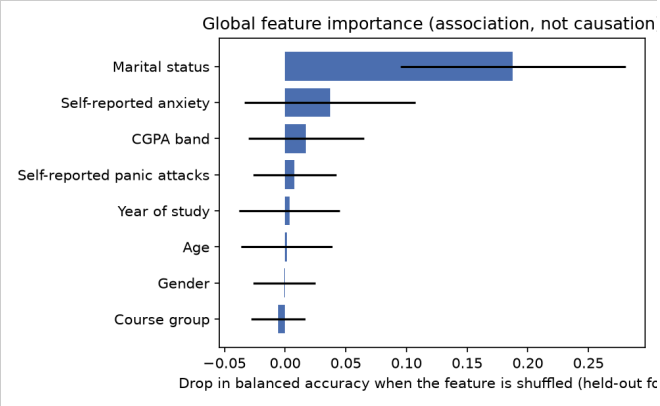


Fig. 3. Global feature permutation importance on held-out test folds.

A crucial contribution of WellSense is exposing why marital status dominates: in this 101-student sample, exactly 16 out of 16 married students answered 'Yes' to depression. The machine learning model captures this 100% conditional distribution as a mathematical shortcut. WellSense explicitly surfaces this finding as a data artifact in the user interface rather than a genuine real-world causal relationship.

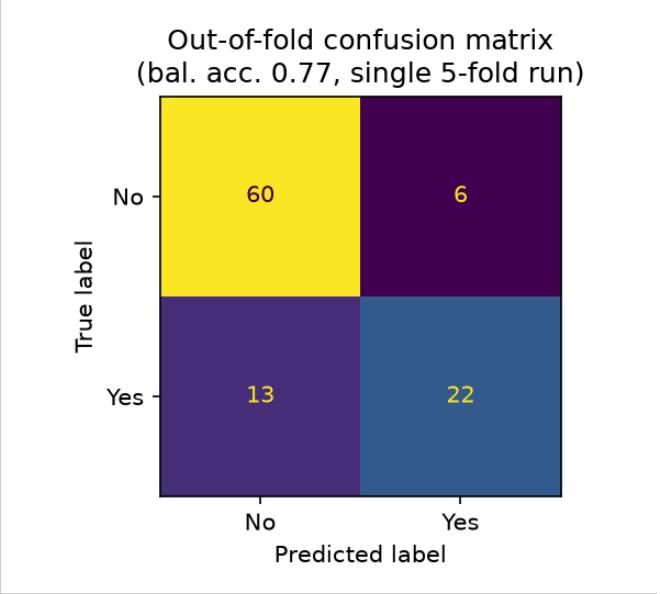


Fig. 4. Out-of-fold confusion matrix for the deployed MLP configuration.

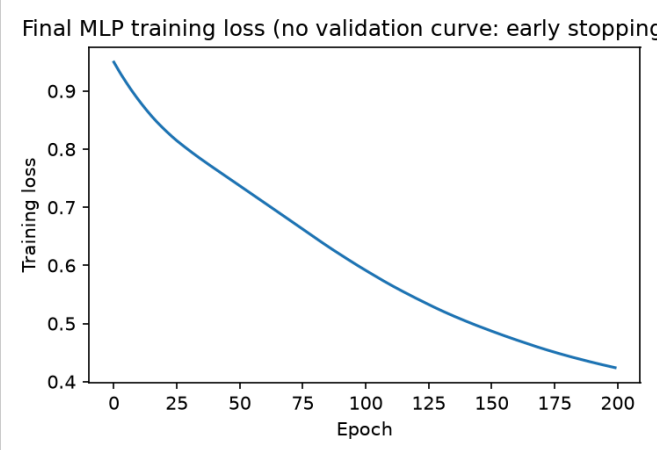


Fig. 5. Final MLP training loss convergence curve.

C. Adaptive Recommendation Dynamics

Table II illustrates closed-loop recommendation re-ranking for a representative student profile before and after receiving negative feedback for sleep suggestions and positive feedback for social connection.

TABLE II. Recommendation Ranking Adaptation Before and After Feedback

Rank	Baseline (No Feedback)	Adapted (After Feedback)
1	Sleep Routine (sleep_routine)	Social Connection (connect)
2	Screen Balance (screen_balance)	Screen Balance (screen_balance)
3	Physical Movement (movement)	Physical Movement (movement)
4	Social Connection (connect)	Sleep Routine (sleep_routine)

V. SYSTEM IMPLEMENTATION & VERIFICATION

WellSense was developed in Python 3.14 using scikit-learn, Streamlit, and SQLite. The web interface comprises six dedicated views: Home, Assessment, Prediction Results, Personalized Recommendations, Feedback & History, and About & Privacy. The entire pipeline is verified by a 30-test unit suite and a headless Streamlit AppTest smoke harness validating standalone page loads, validation boundaries (rejecting age < 16 and > 60), model inference, and SQLite session purging. The full application is open-source and publicly hosted on GitHub at https://github.com/sudheshna03/WellSense.

VI. LIMITATIONS & ETHICAL CONSIDERATIONS

Several limitations govern our findings: (1) Sample Size: The evaluation is based on 101 records; (2) Self-Report Construct: The target reflects a binary questionnaire item rather than a clinical evaluation; (3) Demographic Imbalance: The sample is predominantly female (75%) and undergraduate; (4) Absence of Longitudinal Tracking: Mental health fluctuates dynamically over semesters; and (5) Non-Clinical Scope: WellSense is strictly an educational reflection tool with non-clinical disclaimers prominently maintained across all views.

VII. CONCLUSION

This paper presented WellSense, an explainable, privacy-preserving, and feedback-adaptive framework extending machine learning baselines for student wellbeing support. Applying leak-free repeated nested cross-validation established that a tuned MLP (balanced accuracy: 0.735 ± 0.111) is statistically on par with Random Forest (0.739 ± 0.100), demonstrating that prior claims of 95.6% accuracy are non-reproducible on the public survey cohort. WellSense bridges the gap between passive risk prediction and proactive support through local feature explanations, closed-loop habit recommendations, and a local privacy-by-design architecture. Future work will expand evaluations to multi-institutional cohorts and conduct prospective longitudinal user studies.

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